On the Non-Elliptic Constraint Structure of General Relativity

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Abstract

The non-elliptic principal-symbol structure of the Einstein constraint equations admits a realization within a purely Riemannian compatibility framework with internal normal gauge only. A triple $(h, K, \nabla^{\perp})$ determines a canonical metric connection on $TM \oplus N$ whose flatness is equivalent to the Gauss–Codazzi–Ricci system and defines a first-order internal-gauge constraint geometry on an affine space of orthogonal connections. In codimension one, the two Einstein constraint principal-symbol classes coincide with canonical contractions of the compatibility curvature decomposition.

At flat points and for $n \geq 2$, the reduced principal symbol fails injectivity modulo internal normal gauge, and no admissible internal gauge condition restores ellipticity. In frozen flat charts, the mixed sector admits a local Hessian potential description and generates an infinite-dimensional reduced formal tangent of the flat locus.

1 Introduction

Standing background. Fix a smooth compact boundaryless manifold M, a Riemannian metric h on TM, a Euclidean vector bundle $(N, g_N) \to M$, and the Levi–Civita connection ∇^h of h. The background is fixed globally: h is not varied.

Gauge convention. Gauge quotient uses internal normal gauge $\Gamma(\mathrm{SO}(N))$ only; diffeomorphisms are not quotiented.

Purpose and scope. The Einstein constraint equations are non-elliptic at principal-symbol level [8]. The same degeneracy appears in the intrinsic compatibility-flatness system of the canonical orthogonal connection. The degeneracy survives reduction by admissible internal gauge fixing. In codimension one, the corresponding momentum and Hamiltonian principal-symbol classes coincide with contractions of the mixed and Gauss blocks at flat points.

1.1 Main Results

The principal structural theorems are:

(i) codimension-one momentum-symbol reduction and universality (Theorem 6.2 and Corollary 6.3),

(ii) codimension-one Hamiltonian-symbol reduction from Gauss scalar contraction (Theorem 6.6 and Corollary 6.7),

(iii) reduced non-ellipticity modulo internal gauge at flat points and impossibility of admissible elliptic restoration (Theorems 4.9 and 4.11),

(iv) local potential parametrization of frozen mixed kernel and infinite-dimensional reduced tangent (Theorem 5.1 and Theorem 5.5).

At flat points and for $n \geq 2$, the reduced principal symbol of the compatibility flatness system fails injectivity at every nonzero covector, and no admissible internal gauge fixing restores ellipticity. The mixed-sector kernel has dimension k at each such covector (Theorem 4.4).

2 Canonical Connection and GCR Blocks

2.1 Notation and Conventions

Use the standing background fixed above. The bundle $(N, g_N) \to M$ has rank k and metric connection $\nabla^\perp$. Assume M is oriented henceforth, and let $*$ denote the Hodge star induced by h.

Fix

$$R(X,Y)Z := \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z, \tag{1}$$

for any connection ∇ on a vector bundle.

Set

$$E := TM \oplus N, \qquad g_E := h \oplus g_N. \tag{2}$$

A GCR triple is a datum

$$(h, K, \nabla^\perp), \qquad K \in \Gamma(\mathrm{Sym}^2 T^*M \otimes N). \tag{3}$$

For each $X \in TM$, define

$$K_X : TM \to N, \qquad K_X(Y) := K(X,Y), \tag{4}$$

and its $h \oplus g_N$-adjoint

$$K_X^* : N \to TM, \qquad h(K_X^* \nu, Y) = \langle \nu, K_X Y \rangle_N. \tag{5}$$

Equivalently, regard

$$K \in \Omega^1(M, \mathrm{Hom}(TM, N)), \qquad K(X) := K_X, \tag{6}$$

and

$$K^* \in \Omega^1(M, \mathrm{Hom}(N, TM)), \qquad K^*(X) := K_X^*. \tag{7}$$

Define also

$$A \in \Omega^1(M, \mathrm{Hom}(N, TM)), \qquad A(X) := K_X^*. \tag{8}$$

Equivalently, $A_\nu X := K_X^* \nu$.

All bundle-valued wedges are composition wedges:

$$(\alpha \wedge \beta)(X,Y) = \alpha_X \circ \beta_Y - \alpha_Y \circ \beta_X. \tag{9}$$

For homogeneous form degrees p, q, the graded commutator is

$$[\alpha, \beta]_{\mathrm{gr}} := \alpha \wedge \beta - (-1)^{pq} \beta \wedge \alpha, \qquad \alpha \in \Omega^p(M, \mathrm{End}(E)),\ \beta \in \Omega^q(M, \mathrm{End}(E)). \tag{10}$$

Define

$$(K^* \wedge K)(X,Y) := K_X^* \circ K_Y - K_Y^* \circ K_X \in \mathrm{End}(TM), \tag{11}$$

$$(K \wedge K^*)(X,Y) := K_X \circ K_Y^* - K_Y \circ K_X^* \in \mathrm{End}(N). \tag{12}$$

2.2 Notation for induced covariant derivatives

Let D^T denote the exterior covariant derivative on $\mathrm{End}(TM)$ induced by ∇^h. Let $D^\perp$ denote the exterior covariant derivative on $\mathrm{End}(N)$ induced by $\nabla^\perp$. Let

$$\nabla^{\mathrm{Hom}}_X T := \nabla^\perp_X \circ T - T \circ \nabla^h_X, \tag{13}$$

for $T \in \Gamma(\mathrm{Hom}(TM, N))$, and let D^{Hom} be its exterior covariant derivative on $\mathrm{Hom}(TM, N)$-valued forms.

Type summary. The curvature blocks satisfy

$$\mathcal{G} \in \Omega^2(M, \mathrm{End}(TM)), \quad \mathcal{C} \in \Omega^2(M, \mathrm{Hom}(TM, N)), \quad \mathcal{S} \in \Omega^2(M, \mathrm{End}(N)).$$

Theorem 2.1 (Theorem A: Canonical Assembly)**.** *Given a GCR triple $(h, K, \nabla^\perp)$, there exists a unique metric connection $\nabla^\oplus$ on $E = TM \oplus N$ such that for every $X \in TM$:*

(i) the $TM \to TM$ block of $(\nabla^\oplus)_X$ equals ∇^h_X,

(ii) the $N \to N$ block of $(\nabla^\oplus)_X$ equals $\nabla^\perp_X$,

(iii) the $TM \to N$ block of $(\nabla^\oplus)_X$ equals K_X.

Equivalently, the $N \to TM$ block is $-K^*_X$.

Proof. Define

$$\nabla^\oplus_X (Y \oplus \nu) := (\nabla^h_X Y - K^*_X \nu) \oplus (\nabla^\perp_X \nu + K_X Y). \tag{14}$$

Metricity follows from adjointness of K_X and K^*_X:

$$\begin{aligned} X\langle Y_1 \oplus \nu_1, Y_2 \oplus \nu_2\rangle_E = \langle \nabla^h_X Y_1, Y_2\rangle_h + \langle Y_1, \nabla^h_X Y_2\rangle_h \\ + \langle \nabla^\perp_X \nu_1, \nu_2\rangle_N + \langle \nu_1, \nabla^\perp_X \nu_2\rangle_N, \end{aligned}$$

and the mixed terms cancel:

$$-\langle K^*_X \nu_1, Y_2\rangle_h + \langle K_X Y_1, \nu_2\rangle_N - \langle Y_1, K^*_X \nu_2\rangle_h + \langle \nu_1, K_X Y_2\rangle_N = 0.$$

For uniqueness, let $\widetilde{\nabla}$ satisfy (i)–(iii). Write

$$\widetilde{\nabla} = \nabla^h \oplus \nabla^\perp + \Phi, \tag{15}$$

with $\Phi_X \in \mathrm{End}(E)$. Conditions (i)–(iii) force the $TM \to TM$ and $N \to N$ blocks of Φ_X to vanish and the $TM \to N$ block to be K_X. Metricity forces Φ_X skew-adjoint, hence the $N \to TM$ block is $-K^*_X$. Therefore $\widetilde{\nabla} = \nabla^\oplus$. □

Define the block perturbation 1-form

$$\Psi_X := \begin{pmatrix} 0 & -K^*_X \\ K_X & 0 \end{pmatrix}, \qquad \nabla^\oplus = \nabla^h \oplus \nabla^\perp + \Psi. \tag{16}$$

2.3 Curvature Decomposition

Define operator curvature on E by

$$R^{\oplus}(X,Y) := [\nabla^{\oplus}_X, \nabla^{\oplus}_Y] - \nabla^{\oplus}_{[X,Y]} \in \mathrm{End}(E), \tag{17}$$

and define

$$\mathcal{F}^{\oplus}(X,Y) := R^{\oplus}(X,Y) \in \Omega^2(M, \mathfrak{so}(E)). \tag{18}$$

In any adapted orthonormal frame, with connection matrix Ω of $\nabla^{\oplus}$,

$$\mathcal{F}^{\oplus} = d\Omega + \Omega \wedge \Omega. \tag{19}$$

Proposition 2.2 (Explicit curvature blocks)**.** *With respect to $E = TM \oplus N$, let $\mathcal{C}^* \in \Omega^2(M, \mathrm{Hom}(N, TM))$ denote the fiberwise adjoint of $\mathcal{C}$ with respect to h and $\langle \cdot, \cdot \rangle_N$.*

$$\mathcal{F}^{\oplus} = \begin{pmatrix} \mathcal{G} & -\mathcal{C}^* \\ \mathcal{C} & \mathcal{S} \end{pmatrix}, \tag{20}$$

where

$$\mathcal{G} = R^h - (K^* \wedge K) = R^h - (A \wedge K), \tag{21}$$

$$\mathcal{C} = D^{\mathrm{Hom}} K, \tag{22}$$

$$\mathcal{S} = R^{\perp} - (K \wedge K^*) = R^{\perp} - (K \wedge A). \tag{23}$$

Proof. From $\nabla^{\oplus} = \nabla^{\mathrm{prod}} + \Psi$, with $\nabla^{\mathrm{prod}} := \nabla^h \oplus \nabla^{\perp}$,

$$\mathcal{F}^{\oplus} = \mathcal{F}_{\mathrm{prod}} + D^{\mathrm{prod}}\Psi + \Psi \wedge \Psi. \tag{24}$$

The $TM \to N$ block of $\Psi \wedge \Psi$ vanishes because compositions of two off-diagonal maps are diagonal. The $TM \to N$ block of $D^{\mathrm{prod}}\Psi$ is exactly $D^{\mathrm{Hom}}K$, which gives (22).

For $Z \in TM$,

$$\begin{aligned} ((\Psi \wedge \Psi)_{TT}(X,Y))Z &= \Psi_X(\Psi_Y Z) - \Psi_Y(\Psi_X Z) \\ &= -K^*_X(K_Y Z) + K^*_Y(K_X Z) \\ &= -(K^* \wedge K)(X,Y)Z, \end{aligned}$$

yielding (21). For $\nu \in N$,

$$\begin{aligned} ((\Psi \wedge \Psi)_{NN}(X,Y))\nu &= \Psi_X(\Psi_Y \nu) - \Psi_Y(\Psi_X \nu) \\ &= -K_X(K^*_Y \nu) + K_Y(K^*_X \nu) \\ &= -(K \wedge K^*)(X,Y)\nu, \end{aligned}$$

yielding (23). □

Theorem 2.3 (Theorem B: Integrability Equivalence)**.** *For a GCR triple $(h, K, \nabla^{\perp})$, the following are equivalent:*

(i) $\mathcal{F}^{\oplus} = 0$*;*

(ii) the three equations (21)–(23) *vanish, equivalently the Gauss–Codazzi–Ricci system.*

Proof. By Proposition 2.2, the blocks of $\mathcal{F}^{\oplus}$ are exactly $(\mathcal{G}, \mathcal{C}, \mathcal{S})$. Therefore $\mathcal{F}^{\oplus} = 0$ iff each block vanishes. Those block equations are Gauss, Codazzi, and Ricci in invariant form. In any adapted orthonormal frame, these are equivalent to the component Gauss/Codazzi/Ricci identities. □

Proposition 2.4 (Internal gauge equivariance)**.** *The assignment*

$$(h, K, \nabla^{\perp}) \longmapsto \nabla^{\oplus}$$

is equivariant under internal normal gauge $\Gamma(\mathrm{SO}(N))$*. Consequently,* $\mathcal{F}^{\oplus}$ *and its blocks* $(\mathcal{G}, \mathcal{C}, \mathcal{S})$ *transform covariantly.*

Proof. The gauge action is blockwise conjugation on $\nabla^{\perp}$ and tensor action on K. The assembled form $\nabla^{\oplus} = \nabla^{h} \oplus \nabla^{\perp} + \Psi(K)$ is preserved under this action. Curvature transforms by conjugation, and block projections are equivariant. □

2.4 The Bianchi Identity

Fix canonical pairing

$$\langle A, B \rangle_{\mathrm{can}} := -\mathrm{tr}(AB), \qquad A, B \in \mathfrak{so}(E). \tag{25}$$

For $\mathfrak{so}(E)$-valued forms, use

$$\langle \alpha, \beta \rangle := -\mathrm{tr}(\alpha \wedge *\beta), \qquad \|\mathcal{F}^{\oplus}\|^2 := -\mathrm{tr}(\mathcal{F}^{\oplus} \wedge *\mathcal{F}^{\oplus}). \tag{26}$$

This is the pointwise norm induced by $\langle \cdot, \cdot \rangle_{\mathrm{can}}$ and h. The splitting

$$\mathfrak{so}(E) = \mathfrak{so}(TM) \oplus \big(\mathrm{Hom}(TM, N) \oplus \mathrm{Hom}(N, TM)\big) \oplus \mathfrak{so}(N)$$

is orthogonal under $-\mathrm{tr}$, so cross terms vanish. Then the block-orthogonal split gives

$$\|\mathcal{F}^{\oplus}\|^2 = \|\mathcal{G}\|^2 + 2\|\mathcal{C}\|^2 + \|\mathcal{S}\|^2. \tag{27}$$

Theorem 2.5 (Theorem C: Coupled Bianchi System)**.** *The identity* $D^{\oplus}\mathcal{F}^{\oplus} = 0$ *decomposes as*

$$D^{T}\mathcal{G} + [K^{*}, \mathcal{C}] = 0, \tag{28}$$

$$D^{\mathrm{Hom}}\mathcal{C} + K \wedge \mathcal{S} - \mathcal{G} \wedge K = 0, \tag{29}$$

$$D^{\perp}\mathcal{S} + [K, \mathcal{C}^{*}] = 0. \tag{30}$$

Proof. Write

$$\Psi = \begin{pmatrix} 0 & -K^{*} \\ K & 0 \end{pmatrix}, \qquad \mathcal{F}^{\oplus} = \begin{pmatrix} \mathcal{G} & -\mathcal{C}^{*} \\ \mathcal{C} & \mathcal{S} \end{pmatrix}. \tag{31}$$

Using $D^{\oplus} = D^{\mathrm{prod}} + [\Psi, \cdot]_{\mathrm{gr}}$,

$$0 = D^{\oplus}\mathcal{F}^{\oplus} = D^{\mathrm{prod}}\mathcal{F}^{\oplus} + [\Psi, \mathcal{F}^{\oplus}]_{\mathrm{gr}}. \tag{32}$$

Since $\deg \Psi = 1$ and $\deg \mathcal{F}^{\oplus} = 2$, $[\Psi, \mathcal{F}^{\oplus}]_{\mathrm{gr}} = \Psi \wedge \mathcal{F}^{\oplus} - \mathcal{F}^{\oplus} \wedge \Psi$. Block projection of the identity gives: the (TT) block yields (28), the (TN) block yields (29), and the (NN) block yields (30). This is the block form of Bianchi for principal orthogonal connections [3]. □

Remark 2.6 (Classical immersion consequence)**.** If $U \subset M$ is simply connected and $\mathcal{F}^{\oplus}|_{U} = 0$, the fundamental theorem of submanifolds yields a local immersion with prescribed $(K, \nabla^{\perp})$, unique up to rigid motion and parallel normal gauge [2, Vol. II, Ch. VII, §4]. This statement is not used later.

3 Constraint Geometry of Flatness

Fix $E = TM \oplus N$ with orthogonal bundle metric $g_E = h \oplus g_N$ and fixed splitting type.

Definition 3.1 (Connection configuration space)**.** Define

$$\mathcal{A}_\oplus := \{\nabla^\oplus = \nabla^h \oplus \nabla^\perp + \Psi(K) : \ \nabla^\perp \in \Omega^1(M, \mathfrak{so}(N)), \ K \in \Gamma(\mathrm{Sym}^2 T^*M \otimes N)\}.$$

Definition 3.2 (Internal gauge group and infinitesimal action)**.** Let

$$\mathcal{G} := \Gamma(\mathrm{SO}(N)).$$

Its action is

$$\nabla^\perp \mapsto g^{-1}\nabla^\perp g, \qquad K \mapsto g^{-1} \cdot K.$$

For $\phi \in \Gamma(\mathfrak{so}(N))$, define

$$\tilde{\phi} = \begin{pmatrix} 0 & 0 \\ 0 & \phi \end{pmatrix} \in \Gamma(\mathfrak{so}(E)).$$

Infinitesimally,

$$\delta_\phi \nabla^\oplus = D^{\nabla^\oplus} \tilde{\phi}.$$

With the convention $K \mapsto g^{-1} \cdot K$,

$$\delta_\phi(\nabla^\perp) = D^\perp \phi, \qquad \delta_\phi K = -\,\phi \cdot K.$$

Proposition 3.3. *The $\Gamma(\mathrm{SO}(N))$-action preserves $\mathcal{A}_\oplus$.*

Proof. Orthogonal automorphisms preserve metricity, block type, and the symmetry condition on K. □

Definition 3.4 (Constraint operator and constraint locus)**.** Define the gauge-equivariant first-order operator

$$\Phi : \mathcal{A}_\oplus \to \Omega^2(M, \mathfrak{so}(E)), \qquad \Phi(\nabla) := F^\nabla,$$

so $\Phi(\nabla^\oplus) = \mathcal{F}^\oplus$ for the assembled connection $\nabla^\oplus$, and define the constraint locus

$$\mathcal{Z}_{\mathrm{flat}} := \Phi^{-1}(0) \subset \mathcal{A}_\oplus.$$

Definition 3.5 (First-order internal gauge constraint system)**.** An internal gauge first-order constraint system on a connection space is a gauge-equivariant first-order differential operator whose zero set is gauge-invariant.

Remark 3.6. Orthogonal compatibility is the system $(\mathcal{A}_\oplus, \Phi, \mathcal{G})$ with constraint locus $\mathcal{Z}_{\mathrm{flat}}$.

Lemma 3.7 (Gauge equivariance of Φ)**.** *The curvature map $\Phi : \mathcal{A}_\oplus \to \Omega^2(M, \mathfrak{so}(E))$ is $\mathcal{G}$-equivariant:*

$$\Phi(g^{-1} \cdot \nabla) = g^{-1}\Phi(\nabla)g.$$

Proof. For gauge transform $\nabla \mapsto g^{-1}\nabla g$, standard curvature covariance gives

$$F^{g^{-1}\nabla g} = g^{-1}F^\nabla g.$$

□

Theorem 3.8 (Linearization of the constraint operator)**.** *The map Φ is first-order nonlinear, and for $\eta \in T_{\nabla^\oplus}\mathcal{A}_\oplus$,*

$$D\Phi|_{\nabla^\oplus}(\eta) = D^{\nabla^\oplus}\eta.$$

Proof. For a smooth variation $\nabla_t = \nabla^\oplus + t\eta$, the curvature variation identity gives

$$\frac{d}{dt}\Big|_{t=0} F^{\nabla_t} = D^{\nabla^\oplus}\eta.$$

Hence the Fréchet derivative is $D^{\nabla^\oplus}\eta$. □

Bridge to codimension-one symbols. At flat codimension-one points, contractions of the mixed and Gauss blocks recover the momentum and Hamiltonian principal-symbol classes. Section 6 gives the explicit identifications.

4 Symbol and Reduced Symbol at Flat Points

4.1 The Principal Symbol

Define

$$\mathcal{D}(h, K, \nabla^\perp) := (\mathcal{G}, \mathcal{C}, \mathcal{S}). \tag{33}$$

Let $\mathbf{L}_{(h,K,\nabla^\perp)} := D\mathcal{D}|_{(h,K,\nabla^\perp)}$. Set $\mathbf{L}_{\text{full}} := D\Phi|_{\nabla^\oplus}$ and define the order-zero bundle embedding

$$\iota : \Omega^1(M, \mathfrak{so}(N)) \oplus \Gamma(\mathrm{Sym}^2 T^*M \otimes N) \hookrightarrow \Omega^1(M, \mathfrak{so}(E))$$

by

$$\iota(\delta\nabla^\perp, \delta K)_X = \begin{pmatrix} 0 & -(\delta K)^*_X \\ (\delta K)_X & (\delta\nabla^\perp)_X \end{pmatrix}.$$

Set $\eta := \iota(\delta\nabla^\perp, \delta K)$.

At frozen background $(h, 0, \nabla^\perp)$ with $a = \delta\nabla^\perp = 0$, the remaining block is

$$L(\delta K) := D^{\mathrm{Hom}}(\delta K). \tag{34}$$

In components,

$$(\mathcal{C})_{ij}{}^\alpha{}_k = \nabla_i K^\alpha_{jk} - \nabla_j K^\alpha_{ik}, \tag{35}$$

so principal symbol is

$$(\sigma_\xi \delta K)_{ij}{}^\alpha{}_k = \xi_i\, \delta K^\alpha_{jk} - \xi_j\, \delta K^\alpha_{ik}. \tag{36}$$

Reduced ellipticity means injectivity of the principal symbol after quotienting the domain by the principal-symbol image of internal infinitesimal gauge directions.

Definition 4.1 (Flat point and frozen flat chart)**.** A *flat point* is a connection $\nabla^\oplus \in \mathcal{Z}_{\text{flat}}$, equivalently $F^{\nabla^\oplus} = 0$. A *frozen flat chart at a point x* is a choice of local coordinates and normal-bundle gauge such that the Levi–Civita coefficients of h and the normal-connection coefficients of $\nabla^\perp$ vanish at x. On a simply connected open set U, *frozen flat on U* means these coefficients vanish identically on U.

Lemma 4.2 (Compatibility-chart linearization and principal symbol)**.** *Let $\eta = \iota(\delta\nabla^{\perp}, \delta K)$. Then*

$$\mathrm{pr}_{TT,TN,NN}\big(D\Phi|_{\nabla^{\oplus}}(\eta)\big) = D\mathcal{D}|_{(h,K,\nabla^{\perp})}(\delta\nabla^{\perp}, \delta K),$$

and the principal symbol of $\mathbf{L}_{\mathrm{full}} = D\Phi|_{\nabla^{\oplus}}$ is

$$\sigma_{\xi}(\mathbf{L}_{\mathrm{full}})(\eta) = \xi \wedge \eta.$$

Proof. By Theorem 3.8, $D\Phi|_{\nabla^{\oplus}}(\eta) = D^{\nabla^{\oplus}}\eta$. The first-order term of $D^{\nabla^{\oplus}}$ is exterior differentiation. Block projection then gives the displayed principal parts. □

Lemma 4.3 (Internal gauge principal symbol)**.** *For $\phi \in \Gamma(\mathfrak{so}(N))$ with block embedding*

$$\tilde{\phi} = \begin{pmatrix} 0 & 0 \\ 0 & \phi \end{pmatrix},$$

the infinitesimal gauge action is $\delta_{\phi}\nabla^{\oplus} = D^{\nabla^{\oplus}}\tilde{\phi}$, and its principal symbol is

$$\sigma_{\xi}(\delta_{\phi}\nabla^{\oplus}) = \xi \otimes \tilde{\phi}.$$

At frozen flat background $K = 0$, the principal contribution to the δK chart slot is zero.

Proof. The principal symbol of $D^{\nabla^{\oplus}}$ on 0-forms is $\phi \mapsto \xi \otimes \phi$, hence

$$\sigma_{\xi}(\delta_{\phi}\nabla^{\oplus}) = \xi \otimes \tilde{\phi}.$$

Since $\tilde{\phi}$ is block-diagonal, its principal symbol image has zero off-diagonal block. The δK chart slot is off-diagonal, so the projected principal contribution to δK is zero at frozen $K = 0$. □

Theorem 4.4 (Theorem E: Intrinsic Non-Ellipticity)**.** *Let $n \geq 2$. For every nonzero $\xi \in T_x^*M$,*

$$\sigma_{\xi}(L) : \mathrm{Sym}^2(T_x^*M) \otimes N_x \longrightarrow \Lambda^2(T_x^*M) \otimes T_x^*M \otimes N_x \tag{37}$$

has

$$\dim \ker \sigma_{\xi}(L) = k. \tag{38}$$

At frozen background $K = 0$, infinitesimal gauge action does not contribute to the principal symbol of L in the δK slot. Hence these kernel directions survive principal gauge reduction, and reduced ellipticity fails.

Proof. Fix $\xi \neq 0$. By linear change of basis, take $\xi = e^1$. Then

$$(\sigma_{\xi}\delta K)_{ij}{}^{\alpha}{}_{k} = \delta_{i1}\delta K^{\alpha}_{jk} - \delta_{j1}\delta K^{\alpha}_{ik}. \tag{39}$$

For $i = 1 < j$, equations give $\delta K^{\alpha}_{jk} = 0$ for every $j > 1$, all k, all α. Symmetry $\delta K^{\alpha}_{jk} = \delta K^{\alpha}_{kj}$ forces all components with at least one index > 1 to vanish. Only δK^{α}_{11} remains free. Therefore nullity is one per normal index, total k.

For gauge reduction at $K = 0$, infinitesimal action on K is

$$\delta_{\phi}K = -\,\phi \cdot K, \tag{40}$$

which is linear in K. At $K = 0$, this action vanishes in the δK slot, so it does not contribute to the principal symbol of L in that slot. Thus

$$\mathrm{im}\, \sigma_{\xi}(\delta_{\mathrm{gauge}}) \cap \ker \sigma_{\xi}(L) = \{0\} \tag{41}$$

in the δK sector, and the k-dimensional kernel survives in the reduced symbol. Since $\sigma_\xi(L)$ transforms by bundle isomorphisms under basis change, kernel dimension is basis-invariant; therefore the computation at $\xi = e^1$ is general. Invariantly, the surviving kernel directions are the rank-one tensor classes

$$\delta K = \xi \otimes \xi \otimes \nu, \qquad \nu \in N_x.$$

□

4.2 Analytic Consequences

Proposition 4.5 (Frozen Codazzi non-ellipticity)**.** *For $n \geq 2$, the frozen operator*

$$L = D^{\mathrm{Hom}} : \Gamma(\mathrm{Sym}^2 T^*M \otimes N) \to \Gamma(\Lambda^2 T^*M \otimes T^*M \otimes N)$$

is non-elliptic.

Proof. Theorem 4.4 gives $\dim \ker \sigma_\xi(L) = k$ for every nonzero ξ. Therefore $\sigma_\xi(L)$ is never injective for $\xi \neq 0$, so ellipticity fails. □

Theorem 4.6 (Theorem E$^+$: Non-Fredholm Codazzi Linearization)**.** *Let M^n be compact, smooth, and without boundary, with $n \geq 2$. For every $s \in \mathbb{R}$, the frozen Codazzi operator*

$$L : H^s(\mathrm{Sym}^2 T^*M \otimes N) \to H^{s-1}(\Lambda^2 T^*M \otimes T^*M \otimes N)$$

is not Fredholm.

Proof. By Proposition 4.5, this frozen linear differential operator between fixed bundles is non-elliptic. On closed manifolds, ellipticity is equivalent to existence of a pseudodifferential parametrix, hence equivalent to Fredholmness on Sobolev scales [6, 5, 7, Vol. III, Ch. 20]. Therefore L is not Fredholm. This application is for linear differential operators between fixed vector bundles on compact boundaryless manifolds. □

4.3 The Reduced Symbol

Define the frozen infinitesimal data bundle

$$\mathcal{U} := (\mathrm{Sym}^2 T^*M \otimes N) \ \oplus \ (T^*M \otimes \mathfrak{so}(N)),$$

the defect target bundle

$$\mathcal{V} := (\Lambda^2 T^*M \otimes \mathrm{End}(TM)) \ \oplus \ (\Lambda^2 T^*M \otimes T^*M \otimes N) \ \oplus \ (\Lambda^2 T^*M \otimes \mathfrak{so}(N)),$$

and the gauge-parameter bundle

$$\mathcal{P} := \mathfrak{so}(N).$$

At background $\nabla^\oplus \in \mathcal{A}_\oplus$, set

$$\mathbf{L}_{\mathrm{full}} := D\Phi|_{\nabla^\oplus} : \Gamma(\mathcal{U}) \to \Gamma(\mathcal{V}), \qquad \mathbf{G} := D(\text{gauge action})|_{\nabla^\oplus} : \Gamma(\mathcal{P}) \to \Gamma(\mathcal{U}).$$

For $\xi \neq 0$, reduced ellipticity means injectivity of

$$\bar{\sigma}_\xi(\mathbf{L}_{\mathrm{full}}) : \mathcal{U}_x / \operatorname{im} \sigma_\xi(\mathbf{G}) \to \mathcal{V}_x.$$

Lemma 4.7 (Kernel witness in the mixed sector)**.** *Let $n \geq 2$ and $\xi \neq 0$. There exists nonzero $\delta K \in \mathrm{Sym}^2(T_x^*M) \otimes N_x$ with*

$$\sigma_\xi(L)(\delta K) = 0.$$

Proof. This is the kernel statement proved in Theorem 4.4: $\dim \ker \sigma_\xi(L) = k > 0$ for every $\xi \neq 0$. □

Lemma 4.8 (Gauge-image exclusion for frozen mixed slot)**.** *At frozen flat background, the kernel witness classes from Lemma 4.7 are not in* $\operatorname{im} \sigma_\xi(\mathbf{G})$.

Proof. By Lemma 4.3, the principal gauge symbol has zero projection to the δK slot at frozen $K = 0$. Therefore nonzero δK classes cannot lie in $\operatorname{im} \sigma_\xi(\mathbf{G})$. □

Theorem 4.9 (Theorem E$^\sharp$: Structural Non-Ellipticity Modulo Gauge)**.** *In the background-fixed internal-gauge setting with $\mathcal{G} = \Gamma(\mathrm{SO}(N))$, at each flat point $\nabla^\oplus \in \mathcal{Z}_{\mathrm{flat}}$ and for every nonzero covector ξ, the reduced principal symbol of $(\mathcal{A}_\oplus, \Phi, \mathcal{G})$ fails injectivity for $n \geq 2$.*

Proof. Embed the K-sector

$$U_K = \mathrm{Sym}^2(T_x^*M) \otimes N_x \hookrightarrow \mathcal{U}_x, \qquad \delta K \mapsto (\delta K, 0).$$

By Lemma 4.7, choose $\delta K \neq 0$ with $\sigma_\xi(L)(\delta K) = 0$. By Lemma 4.2, $(\delta K, 0)$ is a nonzero kernel class of $\sigma_\xi(\mathbf{L}_{\mathrm{full}})$. By Lemma 4.8, this class is not removed by quotienting by $\operatorname{im} \sigma_\xi(\mathbf{G})$. Hence $\bar{\sigma}_\xi(\mathbf{L}_{\mathrm{full}})$ is non-injective. □

Definition 4.10 (Admissible internal gauge fixing at flat points)**.** An admissible internal gauge fixing in the background-fixed internal-gauge setting is a first-order operator

$$\mathcal{Q} : T_{\nabla^\oplus} \mathcal{A}_\oplus \to \Gamma(W)$$

whose principal symbol depends only on the normal-connection slot and has zero principal action on the mixed δK slot at flat points.

Theorem 4.11 (No restoration of ellipticity by admissible internal gauge fixing)**.** *Let $\nabla^\oplus \in \mathcal{Z}_{\mathrm{flat}}$ and $n \geq 2$. For any admissible internal gauge fixing $\mathcal{Q}$, the augmented principal symbol*

$$\sigma_\xi(\mathbf{L}_{\mathrm{full}} \oplus \mathcal{Q}) : \mathcal{U}_x / \operatorname{im} \sigma_\xi(\mathbf{G}) \to \mathcal{V}_x \oplus W_x$$

is non-injective for every $\xi \neq 0$.

Proof. By Theorem 4.9, there exists a nonzero class represented by $(\delta K, 0) \in \mathcal{U}_x$ in the kernel of the reduced symbol of $\mathbf{L}_{\mathrm{full}}$. By admissibility, $\sigma_\xi(\mathcal{Q})$ vanishes on the mixed slot at flat points, so the same class is also in the kernel of the reduced augmented symbol. Hence injectivity cannot be restored. □

Corollary 4.12 (No elliptic compatibility sequence at flat points)**.** *The principal-symbol sequence*

$$\Gamma(\mathcal{P}) \xrightarrow{\mathbf{G}} \Gamma(\mathcal{U}) \xrightarrow{\mathbf{L}_{\mathrm{full}}} \Gamma(\mathcal{V})$$

is not elliptic at flat points.

Proof. Ellipticity of the sequence requires injectivity of the reduced symbol of $\mathbf{L}_{\mathrm{full}}$ at every nonzero covector. Theorem 4.9 gives failure of that injectivity. □

Proposition 4.13 (Reduced-symbol coercivity obstruction at flat points)**.** *At flat points and for $n \geq 2$, there is no constant $c > 0$ such that*

$$|\bar{\sigma}_\xi(\mathbf{L}_{\text{full}})u| \geq c|u|$$

for all nonzero ξ and all reduced symbol classes $u \in \mathcal{U}_x / \operatorname{im} \sigma_\xi(\mathbf{G})$.

Proof. By Theorem 4.9, for each nonzero ξ there exists a nonzero reduced class u with $\bar{\sigma}_\xi(\mathbf{L}_{\text{full}})u = 0$. Hence any uniform coercive lower bound fails. □

5 Frozen Kernel and Deformation Consequences

Theorem 5.1 (Theorem E^{++}: Local Potential Parametrization of Frozen Kernel)**.** *Let U be simply connected and let $(U, \nabla^\oplus)$ be a frozen flat chart. Then*

$$\ker(D^{\text{Hom}}) \cap \Gamma(\operatorname{Sym}^2 T^*U \otimes N) = \operatorname{Im}\big(\text{Hess} : C^\infty(U, N) \to \Gamma(\operatorname{Sym}^2 T^*U \otimes N)\big)\,.$$

Equivalently, every frozen mixed-kernel variation is locally Hessian-generated.

Proof. Fix indices (k, α). In frozen flat coordinates, the kernel equation

$$\partial_i \delta K^\alpha_{jk} - \partial_j \delta K^\alpha_{ik} = 0$$

implies the 1-form $\sum_i \delta K^\alpha_{ik}\, dx^i$ is closed. As U is simply connected, the Poincaré lemma ensures the existence of functions V^α_k such that

$$\delta K^\alpha_{ik} = \partial_i V^\alpha_k.$$

The symmetry constraint $\delta K^\alpha_{ik} = \delta K^\alpha_{ki}$ then requires

$$\partial_i V^\alpha_k = \partial_k V^\alpha_i.$$

Applying the Poincaré lemma a second time yields potentials u^α satisfying $V^\alpha_i = \partial_i u^\alpha$, which implies

$$\delta K^\alpha_{ik} = \partial_i \partial_k u^\alpha.$$

The converse inclusion holds trivially by the commutativity of mixed second partial derivatives. We note that the frozen flat hypothesis is strictly necessary; in a general non-flat background, D^{Hom} acquires connection and curvature terms, precluding this exact two-step integration. □

Proposition 5.2 (Local exactness at the frozen mixed stage)**.** *In a frozen flat chart on simply connected U,*

$$0 \to C^\infty(U, N) \xrightarrow{\text{Hess}} \Gamma(\operatorname{Sym}^2 T^*U \otimes N) \xrightarrow{D^{\text{Hom}}} \Gamma(\Lambda^2 T^*U \otimes T^*U \otimes N)$$

is exact at $\Gamma(\operatorname{Sym}^2 T^*U \otimes N)$.

Proof. Exactness at the middle term is exactly Theorem 5.1. □

Lemma 5.3 (Gauge directions tangent to flat locus)**.** *At $\nabla^\oplus \in \mathcal{Z}_{\text{flat}}$, internal gauge directions lie in $\ker D\Phi|_{\nabla^\oplus}$.*

Proof. By Lemma 3.7,

$$\Phi(g^{-1}\cdot\nabla) = g^{-1}\Phi(\nabla)g.$$

Differentiate at $g = \exp(t\phi)$ and $\nabla \in \mathcal{Z}_{\text{flat}}$ with $\Phi(\nabla) = 0$ to obtain

$$D\Phi|_{\nabla}\big(D(\text{gauge})|_{\nabla}\phi\big) = 0.$$

□

Definition 5.4 (Frozen formal tangent and reduced quotient)**.** At $\nabla^{\oplus} \in \mathcal{Z}_{\text{flat}}$, define

$$T_{\nabla^{\oplus}}\mathcal{Z}_{\text{flat}} := \ker\big(D\Phi|_{\nabla^{\oplus}}\big) \subset T_{\nabla^{\oplus}}\mathcal{A}_{\oplus}.$$

Define the reduced frozen formal tangent quotient by

$$T^{\text{red}}_{\nabla^{\oplus}}\mathcal{Z}_{\text{flat}} := T_{\nabla^{\oplus}}\mathcal{Z}_{\text{flat}}\Big/\operatorname{im}\big(D(\text{gauge})|_{\nabla^{\oplus}}\big).$$

Theorem 5.5 (Frozen reduced formal tangent is infinite-dimensional)**.** *Let $n \geq 2$ and let $\nabla^{\oplus} \in \mathcal{Z}_{\text{flat}}$. On each simply connected frozen flat chart U, the reduced tangent quotient $T^{\text{red}}_{\nabla^{\oplus}}\mathcal{Z}_{\text{flat}}$ contains an infinite-dimensional mixed-sector subspace.*

Proof. Work on a simply connected frozen flat chart U.

By Theorem 5.1, for each $u \in C^{\infty}(U, N)$ the section $\delta K = \text{Hess}(u)$ satisfies

$$D^{\text{Hom}}(\delta K) = 0.$$

With $\delta\nabla^{\perp} = 0$, the corresponding variation $\eta = \iota(0, \delta K) \in \Gamma(\Omega^1(U, \mathfrak{so}(E)))$ lies in

$$\ker\big(D\Phi|_{\nabla^{\oplus}}\big) = T_{\nabla^{\oplus}}\mathcal{Z}_{\text{flat}},$$

by Lemma 4.2 and the frozen hypothesis.

Let $\phi \in \Gamma(\mathfrak{so}(N))$ generate an internal gauge direction. In the mixed slot,

$$\delta_{\phi}K = -\,\phi\cdot K.$$

At a frozen flat point one has $K = 0$, hence $\delta_{\phi}K = 0$. Therefore the image of

$$D(\text{gauge})|_{\nabla^{\oplus}} : \Gamma(\mathfrak{so}(N)) \to T_{\nabla^{\oplus}}\mathcal{A}_{\oplus}$$

has zero projection to the mixed δK slot. Consequently, if $\text{Hess}(u) \neq 0$ then the class of $\eta = \iota(0, \text{Hess}(u))$ is nonzero in the quotient

$$T^{\text{red}}_{\nabla^{\oplus}}\mathcal{Z}_{\text{flat}} = T_{\nabla^{\oplus}}\mathcal{Z}_{\text{flat}}/\operatorname{im}\big(D(\text{gauge})|_{\nabla^{\oplus}}\big).$$

Since $\{\,\text{Hess}(u) : u \in C^{\infty}(U, N)\,\}$ is infinite-dimensional, the quotient contains an infinite-dimensional mixed-sector subspace. □

Remark 5.6 (Scope)**.** Theorem 4.9, Theorem 4.11, and Corollary 4.12 are symbol-level structural statements at flat points. No global moduli-dimension claim is asserted.

Corollary 5.7 (Codimension-one frozen deformation consequence)**.** *For $k = 1$ and $n \geq 2$, the internal gauge group is discrete. At flat points, the frozen reduced formal tangent quotient contains an infinite-dimensional mixed-sector subspace generated by Hessian families.*

Proof. For $k = 1$, SO(1) is trivial at infinitesimal level, so internal gauge contributes no nontrivial quotient directions. Apply Theorem 5.5. □

Remark 5.8 (Hypersurface-data interpretation)**.** For codimension one, this statement applies to the frozen compatibility sector of hypersurface data with fixed background metric h: the mixed deformation directions are not elliptically controlled at symbol level.

6 Relation to the Einstein Constraint Symbols

In this section, the scalar Gauss contraction is considered under variations of h in order to identify the principal-symbol classes of the Einstein constraint system. All structural results above remain within the background-fixed internal normal gauge setting.

Definition 6.1 (Codimension-one contraction map)**.** For $k = 1$ in a local unit-normal trivialization, define

$$\mathrm{Contr} : \Lambda^2 T^*M \otimes T^*M \to T^*M, \qquad (\mathrm{Contr}\, C)_i := h^{jk} C_{ij\,k}.$$

Theorem 6.2 (Codimension-one divergence reduction at flat points)**.** *Assume $k = 1$ and work in a frozen flat chart at a flat point. The linearized mixed defect satisfies*

$$(\delta C)_{ij\,k} = \partial_i \delta K_{jk} - \partial_j \delta K_{ik}.$$

After contraction,

$$\delta \bar{C}_i := h^{jk} (\delta C)_{ij\,k} = \partial^j \delta K_{ij} - \partial_i \operatorname{tr}(\delta K).$$

At principal symbol level,

$$\mathrm{Contr} \circ \sigma_\xi(D^{\mathrm{Hom}})(\delta K) = \xi^j \delta K_{ij} - \xi_i \operatorname{tr}(\delta K).$$

Proof. At $k = 1$ the mixed bundle is locally trivialized by a unit normal. In a frozen flat chart, the mixed covariant derivative reduces to partial derivatives at principal order, giving the first formula. Contracting with h^{jk} yields the divergence-trace form. Replacing ∂ by ξ gives the symbol identity. □

Corollary 6.3 (Universality of divergence-symbol degeneracy)**.** *At $k = 1$ and flat points, fix the local unit-normal identification $N|_U \cong U \times \mathbb{R}$, the metric raising map $h^\sharp : T^*U \to TU$, and*

$$\mathrm{Contr}(C)_i = h^{jk} C_{ij\,k}.$$

Then, for each nonzero $\xi \in T^*_x U$, the principal-symbol maps

$$\delta K \longmapsto \mathrm{Contr} \circ \sigma_\xi(D^{\mathrm{Hom}})(\delta K), \qquad \delta K \longmapsto \sigma_\xi\big(\operatorname{div}(\delta K) - d \operatorname{tr}(\delta K)\big),$$

from $\mathrm{Sym}^2 T^*_x U$ to $T^*_x U$ are equal. Thus the two symbols coincide as linear maps $\mathrm{Sym}^2(T^*_x U) \to T^*_x U$ for every nonzero covector ξ.

Proof. Apply Theorem 6.2 and contract with h^{jk} in a unit-normal trivialization of N. □

6.1 Hamiltonian-Side Symbol Identification

Definition 6.4 (Scalar Gauss-contracted compatibility operator)**.** In codimension one, define

$$\mathcal{H}_{\mathrm{comp}}(h, K) := \mathrm{scal}_h - |K|^2_h + (\operatorname{tr}_h K)^2.$$

Remark 6.5. The scalar operator coincides with the hypersurface Hamiltonian constraint: $\mathcal{H}_{\mathrm{GR}} := \mathcal{H}_{\mathrm{comp}}$.

Theorem 6.6 (Codimension-one Hamiltonian symbol comparison at flat points)**.** *Assume $k = 1$ and work at a flat point in a frozen flat chart with $K = 0$. Then*

$$\sigma_\xi\big(D\mathcal{H}_{\mathrm{comp}}\big) = \sigma_\xi\big(\delta\, \mathrm{scal}_h\,\big),$$

and this equals the principal symbol of the linearized Hamiltonian constraint:

$$\sigma_\xi\big(\delta \mathcal{H}_{\mathrm{GR}}\big)(\delta h) = |\xi|^2_h \operatorname{tr}_h(\delta h) - \xi^i \xi^j \delta h_{ij}.$$

Proof. From Proposition 2.2,

$$\mathcal{G} = R^h - (K^* \wedge K).$$

Contracting tangent indices in codimension one gives

$$\mathcal{H}_{\text{comp}} = \text{scal}_h - |K|_h^2 + (\text{tr}_h K)^2.$$

Linearizing at $K = 0$, the K-terms are quadratic and contribute zero at first order, so

$$D\mathcal{H}_{\text{comp}}|_{K=0} = \delta \,\text{scal}_h \,.$$

In a frozen flat chart, with $\delta\Gamma_{ij}^k = \frac{1}{2}(\partial_i \delta h_j{}^k + \partial_j \delta h_i{}^k - \partial^k \delta h_{ij})$, the principal part is

$$(\delta \,\text{scal}_h)_{\text{prin}} = -\partial^i \partial^j \delta h_{ij} + \Delta_h \,\text{tr}_h(\delta h).$$

Replacing ∂ by ξ gives

$$\sigma_\xi(\delta \,\text{scal}_h)(\delta h) = |\xi|_h^2 \,\text{tr}_h(\delta h) - \xi^i \xi^j \delta h_{ij}.$$

Since $\mathcal{H}_{\text{GR}} = \mathcal{H}_{\text{comp}}$, this is also the principal symbol of $\delta\mathcal{H}_{\text{GR}}$ at $K = 0$. □

Corollary 6.7 (Dual universality at flat codimension-one points)**.** *At $k = 1$ and flat points, the momentum and Hamiltonian principal symbols coincide with the symbols of canonical contractions of the mixed and Gauss compatibility blocks:*

$$\begin{aligned} \textit{mixed Codazzi contraction} &\longleftrightarrow \sigma_\xi\big(\,\text{div}\,\delta K - d\ \text{tr}\,\delta K\big), \\ \textit{scalar Gauss contraction} &\longleftrightarrow \sigma_\xi\big(\delta\mathcal{H}_{\text{GR}}\big). \end{aligned}$$

The identification is principal-symbol only; no nonlinear or full-operator equivalence is asserted, and no evolution-system claim is made.

Proof. The momentum-side identification is Corollary 6.3. The Hamiltonian-side identification is Theorem 6.6. Combining both gives the paired symbol-level universality statement. □

Theorem 6.8 (Unified codimension-one constraint-symbol map on $(\delta h, \delta K)$)**.** *Assume $k = 1$ and work at a flat point in a frozen flat chart. Define the joint symbol map*

$$\Sigma_\xi : (\delta h, \delta K) \longmapsto \big(\sigma_\xi\big(\,\text{Contr} \circ D^{\text{Hom}}\big)(\delta K),\ \sigma_\xi\big(D\mathcal{H}_{\text{comp}}\big)(\delta h)\big) \in T_x^* M \oplus \mathbb{R}.$$

Then

$$\Sigma_\xi(\delta h, \delta K) = \big(\sigma_\xi\big(\,\text{div}\,\delta K - d\ \text{tr}\,\delta K\big),\ \sigma_\xi\big(\delta\mathcal{H}_{\text{GR}}\big)(\delta h)\big)$$

for every nonzero covector ξ.

Proof. The first component equality is Corollary 6.3. The second component equality is Theorem 6.6. The joint statement is their product on the direct-sum input slot $(\delta h, \delta K)$. □

For a symmetric two-tensor K on (M, h), define

$$\mathcal{M}(K) := \text{div}\, K - d(\text{tr}\, K).$$

In the frozen flat codimension-one regime, Hessian families $\delta K = \text{Hess}(u)$ from Theorem 5.1 satisfy

$$\sigma_\xi(\mathcal{M})(\delta K)_i = \xi^j \delta K_{ij} - \xi_i \,\text{tr}(\delta K) = \xi^j(\xi_i \xi_j u) - \xi_i(\xi^j \xi_j u) = 0.$$

Thus Hessian variations lie in the kernel of the momentum principal symbol.

By Theorem 6.6, the scalar Gauss contraction yields the Hamiltonian principal symbol at flat codimension-one points with $K = 0$.

Scope boundary. No nonlinear equivalence, no evolution statement, and no diffeomorphism-gauge identification is asserted.

7 Conclusion

The flatness system of the canonical orthogonal connection exhibits the same non-elliptic principal-symbol classes as the Einstein constraint equations within a purely Riemannian framework with internal normal gauge only. At flat codimension-one points, the momentum and Hamiltonian symbol classes coincide with canonical contractions of the mixed and Gauss curvature blocks, and the reduced symbol fails injectivity by a mechanism intrinsic to the compatibility geometry.

The frozen deformation space is infinite-dimensional, parametrized by Hessian potentials, and no admissible internal gauge condition restores ellipticity. All statements are local and at principal-symbol level. Whether the kernel persists under small deformations away from flat points, and whether a global version of the reduced-symbol structure admits a moduli interpretation, remain open.

A Appendix A: Canonical Quadratic Penalties and Mixed-Sector Coupling

Constraint-geometry interface. In this appendix, quadratic invariants are recorded as canonical local penalty densities for constraint violation $\Phi(\nabla) \neq 0$. The block-rigidity statement means any canonical quadratic penalty detects the mixed defect sector with fixed coefficient, so mixed coupling cannot be removed within the canonical quadratic class.

A.1 Quadratic class

Fix

$$\langle A, B\rangle_{\mathrm{can}} := -\mathrm{tr}(AB) \tag{42}$$

on $\mathfrak{so}(E)$. Work with local densities, locally modulo exact forms:

$$\rho, \rho' \in \Omega^n(M), \qquad \rho \sim \rho' \iff \rho - \rho' = d\beta \text{ for some } \beta \in \Omega^{n-1}(M). \tag{43}$$

Theorem A.1 (Theorem G: Quadratic Minimality)**.** *In curvature degree two, with canonical pairing fixed and locally modulo exact forms, the unique nontrivial quadratic invariant density is*

$$-\,\mathrm{tr}(\mathcal{F}^{\oplus} \wedge *\mathcal{F}^{\oplus}) \tag{44}$$

up to an overall constant factor.

Proof. The linear candidate $\mathrm{tr}(\mathcal{F}^{\oplus})$ vanishes identically because $\mathcal{F}^{\oplus}$ is skew-valued. Degree filtration then shows the first nonzero invariant order is quadratic. In the fixed-pairing category, every quadratic local invariant is a scalar multiple of the canonical contraction induced by $-\mathrm{tr}$; locally modulo exact forms this is represented by $-\,\mathrm{tr}(\mathcal{F}^{\oplus} \wedge *\mathcal{F}^{\oplus})$ (Chern–Weil/invariant-polynomial framework [2, 1]). Exact-form ambiguity is quotiented by the stated equivalence relation. □

Remark A.2 (Semisimple caveat)**.** For semisimple non-simple algebras, the space of invariant symmetric bilinear forms can be multidimensional. The theorem fixes the canonical pairing as structural input.

Remark A.3 (Topological caveat)**.** In even dimensions, Chern–Weil representatives (e.g. Pontryagin-type densities) can appear globally as closed forms. Theorem A.1 is a local statement modulo exact forms in the fixed canonical pairing class.

A.2 Quadratic Irreducibility

Definition A.4 (Quadratic decoupling)**.** A quadratic invariant density in the canonical class is *decoupled* if, after block decomposition

$$\mathcal{F}^{\oplus} = \begin{pmatrix} \mathcal{G} & -\mathcal{C}^* \\ \mathcal{C} & \mathcal{S} \end{pmatrix}, \tag{45}$$

it is pointwise equal to a density depending only on $(\mathcal{G}, \mathcal{S})$ and not on $\mathcal{C}$.

Theorem A.5 (Theorem H: Quadratic Constraint Obstruction)**.** *Any quadratic gauge-invariant density built from the fixed canonical pairing is a constant multiple of* $-\mathrm{tr}(\mathcal{F}^{\oplus} \wedge *\mathcal{F}^{\oplus})$*. In block form, the mixed sector enters with fixed coefficient* 2*:*

$$\|\mathcal{F}^{\oplus}\|^2 = \|\mathcal{G}\|^2 + 2\|\mathcal{C}\|^2 + \|\mathcal{S}\|^2. \tag{46}$$

Therefore no nontrivial quadratic invariant suppresses mixed dependence unless $\mathcal{C} \equiv 0$*.*

Proof. Theorem A.1 gives uniqueness of the quadratic invariant class up to scalar. The displayed norm identity is its block expansion under canonical orthogonality. The mixed term is $2\|\mathcal{C}\|^2$ with fixed positive coefficient. Hence suppression of mixed dependence is impossible unless $\mathcal{C} \equiv 0$. □

References

[1] I. Kolář, P. W. Michor, and J. Slovák, *Natural Operations in Differential Geometry*, Springer, 1993.

[2] S. Kobayashi and K. Nomizu, *Foundations of Differential Geometry, Vol. II*, Wiley-Interscience, 1969.

[3] S. Kobayashi and K. Nomizu, *Foundations of Differential Geometry, Vol. I*, Wiley-Interscience, 1963.

[4] R. W. Sharpe, *Differential Geometry: Cartan's Generalization of Klein's Erlangen Program*, Springer, 1997.

[5] L. Hörmander, *The Analysis of Linear Partial Differential Operators I*, Springer, 2nd ed., 1990.

[6] M. E. Taylor, *Partial Differential Equations III: Nonlinear Equations*, Springer, 2nd ed., 2011.

[7] M. A. Shubin, *Pseudodifferential Operators and Spectral Theory*, Springer, 2nd ed., 2001.

[8] R. Bartnik and J. Isenberg, "The Constraint Equations," in *The Einstein Equations and the Large Scale Behavior of Gravitational Fields*, Birkhäuser, 2004, pp. 1–38.

[9] M. Spivak, *A Comprehensive Introduction to Differential Geometry, Vol. IV*, Publish or Perish, 3rd ed., 1999.